

Research and Applications

A study of calibration as a measurement of trustworthiness of large language models in biomedical natural language processing

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Abstract

Objectives: To assess the calibration of 9 large language models (LLMs) within biomedical natural language processing (BioNLP) tasks, furthering understanding of trustworthiness and reliability in real-world settings.

Materials and Methods: For each LLM, we collected responses and corresponding confidence scores for all 13 datasets (grouped into 6 tasks) of the Biomedical Language Understanding & Reasoning Benchmark (BLURB). Confidence scores were assigned using 3 strategies: Verbal, Self-consistency, and Hybrid. For evaluation, we introduced Flex-ECE (Flexible Expected Calibration Error), a novel adaptation of ECE that accounts for partial correctness in model responses, allowing for a more realistic assessment of calibration in language-based settings. Two post-hoc calibration techniques— isotonic regression and histogram binning— were evaluated.

Results: Across tasks, mean calibration ranged from 23.9% (Population-Intervention-Comparison-Outcome extraction) to 46.6% (Relation Extraction). Across LLMs, Medicine-Llama3-8B had the best mean overall calibration (29.8%), and Flan-T5-XXL had the highest ranking on 5/13 datasets. Across strategies, Self-consistency (mean: 27.3%) had better calibration than Verbal (mean: 42.0%) and Hybrid (mean: 44.2%). Post-hoc methods substantially improved calibration, with best mean calibrated Flex-ECEs ranging from 0.1% to 4.1%.

Discussion: The poor out-of-the-box calibration of LLMs poses a risk to trustworthy deployment of such models in real-world BioNLP applications. Calibration can be improved post-hoc and is a recommended practice. Non-binary metrics for LLM evaluation such as Flex-ECE provide a more realistic assessment of trustworthiness of LLMs, and indeed any model that can be partially right/wrong.

Conclusion: This study shows that out-of-the-box calibration of LLMs is very poor, but traditional post-hoc calibration techniques are useful to calibrate LLMs.

Lay Summary

Large-language models (LLMs) are increasingly being used within biomedical research, yet quantitative benchmarking of the trustworthiness and reliability of LLMs is often not performed. This is a risk, especially in healthcare when the output of these models is used to guide decision making. Confidence scores are metrics that can assess the trustworthiness, or reliability, of an LLM's output. For calibration scores to be reliable, the model must be well-calibrated, meaning that the model's confidence and accuracy are broadly aligned. A poorly calibrated model may provide inaccurate responses with high confidence, leading to downstream errors and potentially significant risk to healthcare users.

This study assessed the calibration of 9 LLMs across 13 datasets critical for biomedical research. There was substantial variation in calibration across LLMs; however, even the most calibrated LLMs were approximately 30% off target, indicating the LLMs are poorly calibrated for biomedical tasks. Luckily, traditional and inexpensive correction methods were demonstrated to improve the LLMs' calibration.

This study demonstrates a tailored strategy matching LLMs with biomedical tasks and applying appropriate corrections is required to achieve optimal calibration, and is essential for decision-making in a healthcare setting; a "one-size fits all" approach is not sufficient.

Key words: large language models (LLMs); Biomedical Language Understanding & Reasoning Benchmark (BLURB); calibration, natural language processing (NLP); trustworthiness.

Background and significance

The increasing deployment of large language models (LLMs) in high-stakes domains such as healthcare, where precise and

reliable information is critical, has prompted growing scrutiny over their trustworthiness.¹ Biomedical Natural Language Processing (BioNLP) applications, which encompass

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tasks such as disease identification, medical decision support, and information extraction, require human oversight for interpretation.^{2–4} Confidence metrics can enhance this oversight by serving as proxies for trustworthiness, by associating a response with a probability of the validity of the response.

However, confidence metrics are only reliable if the model is well-calibrated. Calibration is assessed by measuring the degree of alignment between the model's predicted probabilities and true probabilities (eg, accuracy) over a test set.⁵ For example, if a model predicts a 75% chance of cancer in radiographs, we expect 75% of those images to indicate cancer. Calibration ultimately informs developers and users of a model whether predicted probabilities can be interpreted as *confidence* scores.

Model probabilities as confidence scores are useful for downstream decisions and are necessary in high-stakes environments, such as in healthcare. For instance, if a model is well-calibrated, we might trust its predictions without human intervention when confidence scores are extreme (eg, $\geq 95\%$ or $\leq 5\%$), while opting to manually verify predictions with confidence scores outside of the ranges. When a model is *miscalibrated*, we cannot trust predicted probabilities and would need to manually review all predictions. Measuring calibration reveals whether the LLM's confidence in its responses is accurate, indicating whether it is overconfident, underconfident, or *just right*, as illustrated in Figure 1.

Despite recent advancements, LLMs frequently struggle with overconfidence in their predictions,⁶ especially in BioNLP. This is largely due to a lack of domain-specific training, as the most popular LLMs are trained on general-domain data sources like social media and Wikipedia.⁷ There is a growing body of research focused on calibration in LLMs, yet there remains a significant gap in applying these findings to BioNLP tasks for healthcare research.^{1,8} One recent study indeed sheds some light onto the calibration of LLMs in BioNLP,⁹ although for a limited number (and type) of LLMs and tasks. Our paper aims to address this gap by first assessing the degree of calibration of various LLMs in various BioNLP tasks, thereby adding more *depth and breadth* to the understanding of how calibrated LLMs are in BioNLP. Having quantified the degree of calibration—or miscalibration—of LLMs in BioNLP, we take the additional step of evaluating the effectiveness of different techniques to mitigate miscalibration.

Objectives

The objectives of the study are to quantify:

- 1) The degree to which LLMs are calibrated for BioNLP tasks.
- 2) The utility of traditional calibration techniques to mitigate miscalibration of LLMs in the BioNLP domain.

Specifically, we wish to address the following research questions (RQs):

RQ1 How calibrated are state-of-the-art LLMs in the BioNLP domain?

RQ1.1 Is there a relationship between calibration and *family* of LLMs?

RQ1.2 Is there a relationship between calibration and *family* of BioNLP tasks?

RQ1.3 Is there a relationship between calibration and confidence assignment strategy?

RQ2 Can post-hoc calibration techniques mitigate LLM miscalibration?

Materials and methods

Figure 2 illustrates how we systematically assessed the calibration of a diverse set of LLMs. The following sections delve into details of our approach.

BLURB benchmarking datasets and healthcare-based tasks

The Biomedical Language Understanding & Reasoning Benchmark (BLURB; <https://microsoft.github.io/BLURB/tasks.html#overview>) is a comprehensive benchmark for BioNLP with 13 publicly available datasets, grouped in 6 tasks. BLURB has been thoroughly described previously,¹⁰ but Figure 3 provides an illustration of each task for ease of reading.

Models and prompting

Our study included a diverse set of 9 LLMs, differing in both size of learnable parameters (stratified into small [≤ 10 billion parameters] and large [> 10 billion parameters]) and domain specificity of the data used to train (or fine-tune) models, resulting in 4 “families” of models:

- **General domain:**
 - **Small:** Zephyr-7B-Beta,¹¹ Llama-3-8B-Instruct¹²
 - **Large:** Flan-T5-XXL,¹³ Yi-1.5-34B-Chat,¹⁴ GPT-3.5-Turbo,¹⁵ GPT-4¹⁵
- **Biomedical domain:**
 - **Small:** Meditron-7B,¹⁶ Medicine-Llama3-8B^{17,18}
 - **Large:** MedLLaMA-13B¹⁹

All the aforementioned models are *generative* LLMs, which means we need to *prompt* them with natural language (text). We submit a unique prompt for each example of each dataset to LLMs by programmatically filling a *prompt template* with the input text of each example. (Prompt templates for all datasets are available in [Supplementary File 1](#).) The dataset-specific prompt templates contain instructions for what the LLM should do for all examples of the given dataset.

Confidence assignment strategies

As mentioned above, we require (*potential*) confidence scores for each response, so that we can compute a model's calibration. Assigning a confidence score to an LLM's response is not as trivial as it is with traditional classifiers, whose responses *are* confidence scores. LLMs are effectively next word predictors, so the only probabilities we know for certain are the probabilities of each word in the response given the previous words, but not of the response as a whole, or of the claim in the response.²⁰ (Modern LLMs operate with “tokens” rather than “words,” where a “token” can be a whole word or just part of a word. For instance, it could be that an LLM models the word “apple” as 1 token but “misplace” as 2 tokens: “mis” + “place.” Tokens can also be non-words such as “new line” or “end of generation.”)

In line with Xiong et al.,⁶ 3 confidence assignment techniques have been explored (confidence assignment strategies formally defined in [Supplementary File 2](#)):

Prompt Template	Input Text	Response
Based on the provided sentence, answer with NO_ENTITIES if you find no mentions of chemical entities, otherwise, answer with a comma-delimited list of the mentions of chemical entities you find. {INPUT TEXT}	Torsade de pointes ventricular tachycardia during low dose intermittent dobutamine treatment in a patient with dilated cardiomyopathy and congestive heart failure.	NO_ENTITIES 100%
	The authors describe the case of a 56-year-old woman with chronic, severe heart failure secondary to dilated cardiomyopathy and absence of significant ventricular arrhythmias who developed QT prolongation and torsade de pointes ventricular tachycardia during one cycle of intermittent low dose (2.5 mcg/kg per min) dobutamine .	dobutamine 66%
	The mechanisms of proarrhythmic effects of Dubutamine are discussed.	Dubutamine 100%

Figure 1. Examples of GPT-3.5 being overconfident (top), under-confident (middle), and “just right” (bottom) when identifying chemicals (BC5-chemical). Highlighted snippets are the chemicals to identify in each text, and percentages are the confidence scores associated with each response, which in this case were assigned using the Self-consistency strategy (more on strategies below). Examples are a courtesy of the U.S. National Library of Medicine and can be accessed at <https://huggingface.co/datasets/bigbio/blurb>.

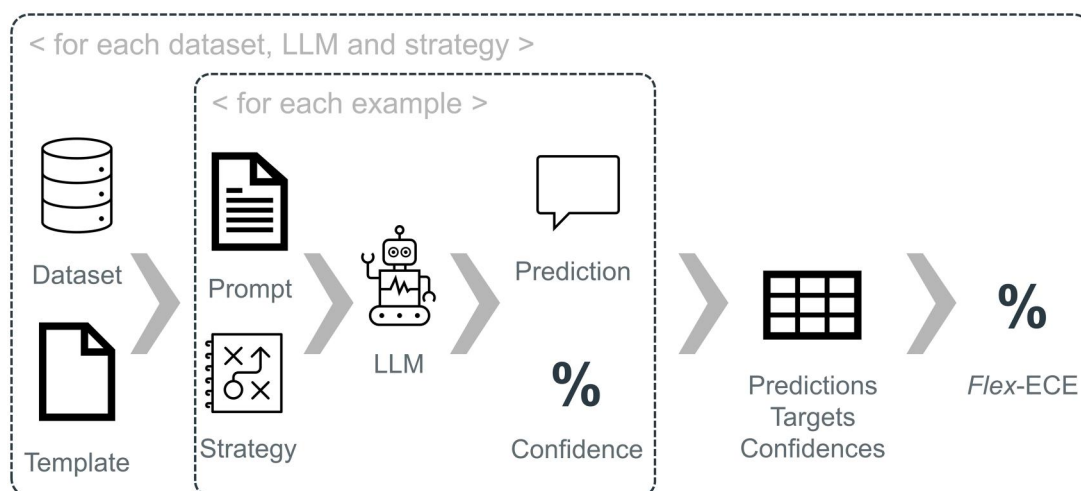


Figure 2. A Flex-ECE score is computed for an LLM with respect to a particular dataset (task) and confidence assignment strategy.

- **Verbal:** Consists of simply including a request in the prompt to return a confidence score in addition to the response; more sophisticated verbal strategies have not proven to yield better confidence scores.^{6,20} While there is no guarantee that verbal confidence aligns with token probabilities of the model’s response, previous research has shown verbal confidence can be better calibrated than token probabilities.²⁰
- **Self-consistency:** Consists of sampling multiple responses for the same prompt, then computing the degree of *similarity* between the several candidate responses and the original response. (To guarantee that the LLM *can* vary responses, we prompt the LLM with an increasing degree of freedom when sampling, by increasing the parameter *temperature* before each call.) The final confidence score is the average of similarity scores.
- **Hybrid:** Consists of combining both above strategies, by eliciting multiple verbal confidence scores, then boosting

or penalizing the initial verbal score depending on the consistency of the responses. It should be noted that *Hybrid* is not guaranteed to be superior to either *Verbal* or *Self-consistency*, since the interplay between these 2 fundamentally distinct approaches could be detrimental to one another, hence the need to empirically verify the effects of combining both strategies.

Evaluation

Traditional calibration metrics such as *Expected Calibration Error* (ECE)⁵ assume that models’ predictions are entirely right or wrong, so that true probabilities become the ratio of right answers for a given test set, ie, *accuracy*. Treating LLM responses as entirely right or wrong is not desirable, often not feasible.²¹

This study thus proposes a *flexibilization* of ECE, dubbed *Flexible ECE* (or *Flex-ECE* for short), where model accuracy is the average *similarity* between generated and target

Task	Input	Target
NER	Immunohistochemical staining was positive for S-100 in all 9 cases stained, positive for HMB-45 in 9 (90%) of 10, and negative for cytokeratin in all 9 cases in which myxoid melanoma remained in the block after previous sections.	S-100 HMB-45 cytokeratin
PICO	Xylitol for prevention of acute otitis media.	P: acute otitis media I: Xylitol C: O:
RE	Cimetidine: Cimetidine increases nicardipine HCl plasma levels.	MECHANISM
SS	Sent-1: It has recently been shown that Craf is essential for Kras G12D-induced NSCLC. Sent-2: It has recently become evident that Craf is essential for the onset of Kras-driven non-small cell lung cancer.	4.0
DC	Biopsy of a skin lesion showed lymphoproliferative infiltration of the dermis with a follicular and angiocentric growth pattern and regional epidermal necrosis.	resisting cell death
QA	Question: Does metformin interfere thyroxine absorption? Context: LT4 absorption is unchanged by concomitant metformin ingestion. It has been hypothesized that metformin may suppress serum thyrotropin (TSH) concentrations by enhancing LT4 absorption or by directly affecting the hypothalamic-pituitary axis.	no

Figure 3. BLURB datasets contain an input text, which we inject in prompt templates, and a target (correct) label, which is used to compute similarity (prediction vs target), and ultimately calibration (confidence vs similarity).

responses for a given test set (Flex-ECE is formally defined in [Supplementary File 2](#)):

$$ECE_{Flex} = \sum_{m=1}^M \frac{|B_m|}{n} |sim(B_m) - conf(B_m)|,$$

where B is a bin of $\langle response, target, confidence \ score \rangle$ tuples, M is the number of bins, n the size of the dataset, sim is the average similarity in the bin, and $conf$ the average confidence in the bin. More formally, sim is defined as follows:

$$sim(B) = \frac{1}{|B_m|} \sum_{i \in B_m} s(y_i, \tilde{y}_i),$$

where y is a response, $\tilde{y}$ is the target response, and s a task-specific similarity function is employed, as illustrated in [Figure 4](#). $conf$ is defined as follows:

$$conf(B) = \frac{1}{|B_m|} \sum_{i \in B_m} c_i,$$

where c is a confidence score. The binning approach of Flex-ECE is the same as in conventional ECE (we use 5 bins for all experiments).

It should be noted that some of the similarity functions differ from the typical metrics used to evaluate certain BLURB tasks. For example, relative distance is used for sentence similarity since Pearson correlation, the usual metric, is unsuitable since it cannot be applied to a single example. We also use Jaccard similarity instead of F1-score to evaluate NER, in line with previous studies²² that have employed similarity scores when measuring the performance of LLMs on named entity recognition. This has the benefit of avoiding the need

to use imperfect heuristics to map predicted entities onto the original example, and more closely aligns with how the model is presented with the predictions in the Verbal confidence scenario. Similarity functions for each task are formally defined in [Supplementary File 2](#).

We highlight that, to the best of our knowledge, Flex-ECE is the first attempt to bring the notion of partial correctness into ECE, although partial correctness as such is not novel in NLP research. Traditional (NLP) metrics such as Levenshtein Distance,²³ BLEU,²⁴ ROUGE,²⁵ all account for partial correctness, but have been traditionally employed to evaluate model correctness in isolation (ie, performance) and not correctness with respect to confidence (ie, calibration). Note that traditional metrics for partial textual correctness can indeed be used as similarity functions for Flex-ECE, provided that the task at hand is one where the prediction and target are both text, and such metrics can quantify single prediction-target pairs (as opposed to only sets of prediction-target pairs, as is the case with Pearson correlation).

Post-hoc calibration

In addition to quantifying the out-of-the-box degree of (mis) calibration of LLMs, we also investigated ways to mitigate miscalibration. We explore only post-hoc calibration techniques, ie, we modify confidence scores only while leaving models intact, since re-training LLMs can be very costly.²⁶ We wish to first understand to what extent traditional, inexpensive calibration techniques can be used to extract more accurate confidence scores from LLMs.

Two techniques were tested:

- **Isotonic regression:** used to learn the rescaling factor that arranges the observed set of probabilities in a stepwise, monotonically increasing line.²⁷

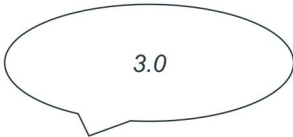
Task / Input	Target	Prediction	Similarity
Single-label NER: Classify the relation between the marked chemical and gene in: <i>While it is an antagonist at these latter receptors, ziprasidone behaves as a 5-HT1A agonist in vitro in adenylate cyclase measurements.</i>	CPR:5		100% (equality)
Multi-label NER: Identify all chemicals in: <i>OBJECTIVE: To study the 2 drugs most clearly implicated (clozapine and olanzapine) and risperidone using a frequently sampled intravenous glucose tolerance test.</i>	clozapine, olanzapine, risperidone, glucose		66% (Jaccard)
Similarity: Rate the similarity (0.0 – 4.0) between: <i>1) Recently, it was reported that expression of IDH1R132H suppresses TET2 activity and the... 2) the mechanism was clarified by yet another genomic survey, this time involving acute...</i>	1.6		70% (Relative Distance)

Figure 4. Task-specific similarity functions prevent us from severely punishing models in scenarios where being partially correct would be acceptable, by allowing correctness to be continuous (eg, a percentage) rather than binary (ie, right or wrong).

- **Histogram binning:** a stricter version of isotonic regression, where the steps (or bins) are pre-defined or are distributed so that each bin has an equal sample size.²⁸

Isotonic regression calibration learns a piecewise monotonically increasing function f and can be defined as follows:

$$\min \sum_{i=1}^n (y_i - f(\hat{p}_i))^2$$

subject to $f(\hat{p}_i) \leq f(\hat{p}_j)$, where y_i is the similarity and $\hat{p}_i$ is the predicted probability for sample i . A histogram binning calibrator can be defined as follows:

$$\min_{\theta_1, \dots, \theta_M} \sum_{m=1}^M \sum_{i=1}^n \mathbf{1}(a_m \leq \hat{p}_i < a_{m+1}) (\theta_m - y_i)^2,$$

where M is the number of bins, $\mathbf{1}$ is the indicator function, a_m and a_{m+1} are the boundaries of bin m , $\hat{p}_i$ is the predicted probability, y_i is the similarity, and the calibrated probability for a probability falling into bin m is set to θ_m . In our experiments, we set M to 10 and use uniform-mass binning to determine the bin boundaries so that all bins have an equal number of examples.

To avoid overfitting, we fit each calibration method on a randomly selected subset of the train split of each BLURB dataset, using the uncalibrated confidences from the best performing confidence elicitation strategy for each model, as determined by mean Flex-ECE across datasets. As some calibration methods (particularly isotonic regression) are prone to overfitting when there is insufficient calibration data,²⁹ we test the effect of varying the number of calibration samples (10, 50, 100, 500, and 1000).

Infrastructure

We ran experiments one LLM at a time and utilized the following hardware to serve each self-hosted LLM (which excludes both GPT models): 8-32 × vCPUs, 32-64GB of RAM, and 1 Nvidia GPU with 32-40GB of RAM for most LLMs, except for the 2 largest models (Yi-1.5-34B-Chat and Flan-T5-XXL), where we used 2 such GPUs. To execute experiment scripts, 2 × vCPUs with 8GB of RAM and no GPUs were used. Scripts were all written in Python 3.9, including our self-developed calibration metric *Flex-ECE*.

Results

The number of examples per task ranged from 20 (Sentence similarity) to 5038 (Gene recognition) (Table 1). A verbal confidence score was extracted (defined as a “success”) for 100% of examples for the majority of dataset-LLM combinations. Exceptions where <50% success was observed were Medicine-Llama3-8B (Biomedical question answering with context task: 17.0% success); Meditron-7B and Zephyr-7B-Beta (all tasks other than PICO extraction: <6.8% success); and MedLLaMA-13B (Chemical recognition: 16.4%; Hallmarks of Cancer: 22.0%). When the Self-consistency and Hybrid strategies were used, all LLMs had 100% success rates, except GPT-3.5 and GPT-4 (≥99.8% across all datasets).

RQ1: How calibrated are state-of-the-art LLMs in the BioNLP domain?

The calibration of LLMs was generally poor for tasks in the biomedical domain (Table 1). The Biomedical question answering without context task had the best calibration (mean: 21.4%), followed by the Sentence similarity task (mean: 23.9%). The Chemical-protein relationship extraction

Table 1. Mean Flex-ECE scores by dataset and model.

NLP task	Medical task	Dataset	Examples ^a	Bio-small		General-small		Bio-large		General-large			Mean [SD]
				M-7	ML-8	L-8	Z-7	ML-13	F-XXL	G-3	G-4	Y-34	
Document classification	Hallmarks of cancer identification	HoC	3547	49.0	35.7	39.2	51.8	26.1	47.6	26.3	39.5	43.4	39.5 [22.2]
Named-entity recognition	Disease recognition	BC5-disease	4797	48.3	27.4	34.3	44.1	30.3	43.7	37.6	26.7	38.0	36.0 [16.3]
		NCBI-disease	940	48.7	24.3	34.4	43.4	30.5	43.4	29.8	22.8	33.3	33.6 [17.0]
	Gene recognition	BC2GM	5038	47.5	23.7	37.9	41.7	50.7	45.9	35.6	31.8	38.1	38.8 [19.2]
	Molecular biology term recognition	JNLPBA	4260	48.1	36.6	49.7	47.2	39.4	35.3	46.2	44.8	57.7	45.0 [19.7]
PICO extraction	Chemical recognition	BC5-chemical	4797	48.8	16.5	18.8	43.7	43.4	40.9	22.9	21.9	20.9	29.6 [20.6]
	Overall ^b	Overall ^b	19832	48.3	25.7	35.0	44.0	38.8	41.8	34.4	29.6	37.6	36.5 [19.0]
	PICO extraction from abstracts	EBM-PICO	187	33.4	35.5	49.0	32.7	32.9	30.7	53.9	56.9	55.1	42.2 [29.4]
Relation extraction	Chemical-protein relationship	ChemProt	3069	49.5	50.0	56.9	41.6	54.2	32.8	55.2	57.1	58.3	51.0 [21.7]
Question answering (QA)	Drug-drug interaction	DDI	790	49.6	51.8	52.8	42.4	42.4	51.6	44.5	42.5	54.6	48.2 [17.0]
	Gene-disease association	GAD	534	50.1	31.4	46.3	41.7	31.0	26.3	47.0	51.0	44.7	40.7 [19.9]
	Overall ^b	Overall ^b	4393	49.7	44.4	52.0	41.9	42.5	36.9	48.9	50.2	52.5	46.6 [19.8]
	Biomedical QA without context	BioASQ	140	49.2	10.3	11.8	47.0	28.0	22.0	13.9	13.8	14.6	21.4 [19.9]
	Biomedical QA with context	PubMedQA	500	49.3	26.7	26.8	49.8	35.7	24.7	41.7	44.2	42.1	37.4 [20.8]
Sentence similarity	Overall ^b	Overall ^b	640	49.3	16.8	19.3	48.4	31.8	23.3	27.8	29.0	28.3	29.2 [21.7]
	BIOSSES	BIOSSES	20	48.6	16.7	16.1	50.0	22.3	32.6	10.8	13.3	21.3	23.9 [26.4]
				47.2 [42.4]	29.8 [21.7]	36.5 [15.4]	44.0 [31.0]	36.1 [23.3]	36.7 [16.7]	35.8 [15.3]	35.9 [15.5]	40.2 [15.6]	37.5 [22.4]
				Mean [SD]									

The calibration scores in this table represent the mean of the Flex-ECE derived from each of the 3 confidence score elicitation strategies. Bold values denote the best performing model(s) at each task, as well as the best mean Flex-ECE across all tasks (rightmost column) and models (bottom row). Only task-model-strategy combinations with a success rate $\geq 50\%$ are included. PICO: Population, Intervention, Comparator, Outcome; QA: Question answering; NLP: Natural language processing; SD: Standard deviation. Models are abbreviated for ease of reading: F-XXL = Flan-T5-XXL; G-3 = GPT-3.5; G-4 = GPT-4; ML-13 = MedLLaMA-13B; ML-8 = Medicine-Llama3-8B; M-7 = Meditron-7B; L-8 = Meta-Llama-3-8B-Instruct; Y-34 = Yi-1.5-34B-Chat; Z-7 = Zephyr-7B-Beta. Bio-small: M7, ML-8; General-Small: Z-7, L-8; Bio-Large: ML-13; General-Large: F-XXL, Y-34, G-3, G-4.

^a The number of examples within each of the datasets (eg, within the HoC dataset, there were 3547 different example texts for LLMs to label with hallmarks of cancer).

^b The Overall rows indicate the mean of the calibration scores from each dataset for a given NLP task. For example, the Overall row of the Named-Entity Recognition section is the mean of the calibration scores from the BC5-disease, NCBI-disease, BC2GM, JNLPBA, and BC5-chemical datasets.

task (mean: 51.0%) had the worst calibration, followed by the Drug-drug interaction extraction task (mean: 48.2%).

Medicine-Llama3-8B had the best calibration across the datasets (mean: 29.8%; Table 1). Five of the LLMs had similar mean calibration scores across the datasets (means ranging from 35.8% to 36.7%; ordered by increasing score: GPT-3.5, GPT-4, MedLLaMA-13B, Meta-Llama-3-8B-Instruct, and Flan-T5-XXL). The LLMs with the worst calibration were Yi-1.5-34B-Chat (mean: 40.2%), Zephyr-7B-Beta (mean: 44.0%), and Meditron-7B (mean: 47.2%). Medicine-Llama3-8B was ranked as the best calibrated model for only 3/13 datasets whereas Flan-T5-XXL was ranked as the best calibrated model for 5/13 datasets.

For the majority of tasks (7/13), multiple LLMs displayed similar calibration scores, ie, no LLM(s) displayed markedly better calibration: (1) Molecular biology term recognition, (2) Chemical recognition, (3) PICO-extraction, (4) Drug-drug interaction, (5) Biomedical QA without context, (6) Biomedical QA with context, and (7) Sentence Similarity tasks. For the remaining tasks (6/13), 1 or 2 models had better calibration than the rest: Hallmarks of cancer (MedLLaMA-13B and GPT-3.5), Disease recognition (GPT-4 and Medicine-Llama3-8B), Gene recognition (Medicine-Llama3-8B), Chemical-protein relationship (Flan-T5-XXL), and Gene-disease association (Flan-T5-XXL) tasks (Table 1).

We also find that standard ECE tends to overestimate the degree of miscalibration when compared to Flex-ECE. For the tasks where the calculation of ECE and Flex-ECE differ (ie, all tasks besides relation extraction and QA, which use the equality similarity function), the mean ECE was greater than Flex-ECE. The difference ranged from 1.5 percentage points for Document classification to as high as 36.5 percentage points for Sentence similarity (see Supplementary File 3 for a comparison of ECE and Flex-ECE across datasets).

In line with Ao et al.,³⁰ we have computed non-absolute Flex-ECE scores in order to quantify the degree to which LLMs are overconfident (ie, *negative* non-absolute Flex-ECE) or underconfident (ie, *positive* non-absolute Flex-ECE). We adopt an arbitrary threshold of $\pm 1\%$ to account for scenarios where the LLM was neither over- nor underconfident, but *just right*, ie, the LLM is well-calibrated for that specific experiment scenario. As depicted in Table 2, we find that in the vast majority of scenarios (296/351 or 84.3%) LLMs are overconfident, with 4/9 LLMs (both GPT models included) being overconfident in all scenarios.

Upon manually inspecting examples where models are either over- or underconfident (as depicted in Figure 5), we observe that there exist essentially 3 classes of errors:

- 1) *Legitimate overconfidence*: when the model's confidence in a valid yet (mostly) wrong answer is high.
- 2) *Legitimate underconfidence*: when the model's confidence in a valid and (mostly) right answer is low.
- 3) *Apparent overconfidence*: when the model displays high confidence in an answer that indeed contains a (mostly) right answer, but also unsolicited output, so the answer is unfortunately treated as (mostly) wrong.

RQ1.1: Is there a relationship between calibration and family of LLMs?

The families of models (Bio-Large; Bio-Small; General-Large; General-Small) had similar mean calibrations across the tasks, ranging from 36.1% (Bio-Large) to 39.5%

Table 2. Per-model distribution of non-absolute Flex-ECE across all 351 experiment scenarios (ie, 39 per model across 13 tasks).

Model	Overconfident	Underconfident	Just right
Flan-T5-XXL	24 (61.5%)	15 (38.5%)	0 (0.0%)
GPT-3.5	39 (100.0%)	0 (0.0%)	0 (0.0%)
GPT-4	39 (100.0%)	0 (0.0%)	0 (0.0%)
MedLLaMA-13B	30 (76.9%)	9 (23.1%)	0 (0.0%)
Medicine-Llama3-8B	30 (76.9%)	7 (17.9%)	2 (5.1%)
Meditron-7B	32 (82.1%)	0 (0.0%)	7 (17.9%)
Meta-Llama-3-8B-Instruct	39 (100.0%)	0 (0.0%)	0 (0.0%)
Yi-1.5-34B-Chat	39 (100.0%)	0 (0.0%)	0 (0.0%)
Zephyr-7B-Beta	24 (61.5%)	7 (17.9%)	8 (20.5%)
Total	296 (84.3%)	38 (10.8%)	17 (4.8%)

We view a model as overconfident model when non-absolute Flex-ECE $< -1\%$, as underconfident when non-absolute Flex-ECE $> 1\%$, and “just-right” otherwise (ie, confidence and accuracy are approximately the same across Flex-ECE bins, thus the model is well-calibrated). The $\pm 1\%$ threshold is arbitrary.

(General-Small) (Table 3). However, there was variation in the mean calibration scores across the families of models depending on NLP task.

The Bio-Large model (ie, MedLLaMa-13B) had the best calibration on the Document classification task, across several of the Named-entity recognition family of tasks, and the Drug-drug interaction and Gene-disease association tasks within the Relation extraction family of tasks. However, the Bio-Large model had the worst calibration for the Chemical recognition and Gene recognition task within the Named-entity recognition family of tasks. The General-Large models had the best calibration on the Biomedical question answering task without context. The large models had better calibration than the small models on the Sentence similarity task. The families of models had relatively similar calibration across the remaining tasks.

There was also substantial variation within these families. For example, the Bio-Small family included the model with the best calibration (Medicine-Llama3-8B) and the worst calibration across the datasets (Meditron-7B).

RQ1.2: Is there a relationship between calibration and family of BioNLP tasks?

Three families of BioNLP tasks contained more than one dataset (“Overall” rows of Table 1). Medicine-Llama3-8B had the best mean calibration scores within the Named-entity recognition family (mean: 25.7%) and the Question answering family (mean: 16.8%). Meditron-7B and Zephyr-7B-Beta had the worst calibration scores for these 2 BioNLP task families; the other LLMs had broadly similar calibration scores. Regarding the Relation extraction family, the LLMs had similarly poor calibration.

RQ1.3: Is there a relationship between calibration and confidence elicitation strategy?

Table 3 shows that the Self-consistency strategy (mean: 27.3%) was associated with the best calibration scores compared with the Verbal (mean: 42.0%) and Hybrid (mean: 44.2%) strategies across all tasks. The Self-consistency strategy had the best calibration scores for 11 of the 13 tasks, with the only exceptions being the Gene-disease association task (best calibration = Hybrid strategy) and the Biomedical question answering without context (Verbal strategy had marginally better calibration). It should be noted that Hybrid

Scenario	Target	Prediction	Scores	
GPT-4 DDI Verbal	DDI-false	DDI-mechanism	100% Sim. = 0% Conf. = 100% $\Delta = -100\%$ (overconf.)	Legitimate
FLAN-XXL BC2GM Self-Consistency	- FN promoter - chloramphenicol acetyltransferase	- FN promoter - chloramphenicol acetyltransferase	40% Sim. = 100% Conf. = 40% $\Delta = 40\%$ (underconf.)	Legitimate
Zephyr-7B-Beta ChemProt Hybrid	CPR:6	CPR:6 Explanation: The given text describes the antagonistic effect of EMP metabolites on the androgen receptor. ...	100% Sim. = 0% Conf. = 100% $\Delta = -100\%$ (overconf.)	Apparent

Figure 5. The different types of error. Some discrepancies between similarity and confidence are legitimate, ie, the generated text is valid, but confidence and similarity (correctness) are not aligned. Other discrepancies are due to the wrong format of generations, for example when LLMs generate more than the solicited answer, but the right answer(s) are contained in the text. When that happens and confidence is high, the final calibration score for this example will point to apparent overconfidence, since the LLM was actually correct (or mostly correct) but was penalized by essentially generating more than it was asked to.

Table 3. Mean Flex-ECE scores by dataset and families of models (ie, whether LLMs are small [$<10B$] or large [$>10B$], and domain-specific [biomedical] or general domain) and confidence elicitation strategy.

NLP task	Medical task	Dataset	Family of model				Confidence elicitation strategy		
			Bio-small	General-small	Bio-large	General-large	Verbal	Self-consistency	Hybrid
Document classification	Hallmarks of cancer identification	HoC	41.0	44.2	26.1	39.2	48.1	22.2	51.2
Named-entity recognition	Disease recognition	BC5-disease	35.8	38.2	30.3	36.5	42.7	26.0	40.6
	Gene recognition	NCBI-disease	34.0	38.0	30.5	32.3	40.1	22.5	39.6
	Molecular biology term recognition	BC2GM	33.2	39.4	50.7	37.9	47.5	24.1	46.7
	Chemical recognition	JNLPBA	41.2	48.7	39.4	46.0	55.4	31.0	52.1
PICO extraction	PICO extraction from abstracts	BC5-chemical	29.5	28.8	43.4	26.6	33.9	18.2	37.8
Relation extraction	Chemical-protein relationship	EBM-PICO	34.5	40.9	32.9	49.2	48.2	22.2	56.4
	Drug-drug interaction	ChemProt	49.8	50.8	54.2	50.8	62.2	37.6	55.8
	Gene-disease association	DDI	50.9	48.6	42.4	48.3	56.0	39.6	50.7
Question answering (QA)	Biomedical QA without context	GAD	38.9	44.5	31.0	42.2	41.3	51.7	29.1
	Biomedical QA with context	BioASQ	25.9	25.9	28.0	16.1	16.4	17.5	29.1
Sentence similarity	Biomedical sentence similarity	PubMedQA	38.0	36.0	35.7	38.2	36.6	29.4	45.8
		BIOSSES	29.4	29.7	22.3	19.5	17.1	13.2	39.7
Mean [SD]			37.0 [32.8]	39.5 [23.1]	36.1 [23.3]	37.1 [15.7]	42.0 [19.2]	27.3 [18.0]	44.2 [25.0]

Only task-model-strategy combinations with a success rate $\geq 50\%$ are included. Bold values denote the best performing model(s) at each task, as well as the best mean Flex-ECE across all tasks (rightmost column) and models (bottom row). Bio-small: Meditron-7B, Medicine-Llama3-8B; General-Small: Zephyr-7B-Beta, Llama-3-8B-Instruct; Bio-Large: MedLLaMa-13B; General-Large: Flan-T5-XXL, Yi-1.5-34B-Chat, GPT-3.5, GPT-4; PICO: Population, Intervention, Comparator, Outcome; QA: Question answering; NLP: Natural language processing; SD: Standard deviation.

^a The calibration scores in the family of models section represent the mean of the Flex-ECE derived from each of the 3 confidence score elicitation strategies.

was better calibrated than Verbal in 7/13 tasks, suggesting that aggregating multiple verbal confidences scores (Hybrid strategy; see details in Section Confidence Assignment Strategies) is not clearly superior to eliciting a single verbal confidence score (Verbal strategy).

However, this pattern was not observed for all LLMs. Four LLMs had better mean calibration scores across tasks when using the Hybrid strategy compared with the Self-consistency strategy (GPT-3.5, GPT-4, Meta-Llama-3-8B-Instruct, Yi-1.5-34B-Chat). For several models (eg, Meditron-7B and Zephyr-7B-Beta), there was a large difference in the mean scores between the different strategies (mean score by strategy—Meditron-7B: Self-consistency = 15.6%, Hybrid = 82.3%; Zephyr-7B-Beta: Self-consistency = 24.0%, and Hybrid = 66.6% [Verbal not considered for these models due to low success rate]), which was less apparent for other models (e.g., Meta-Llama-3-8B-Instruct: Self-consistency = 35.7%, Hybrid = 32.4%, and Verbal = 41.3%).

RQ2: Can post-hoc calibration techniques mitigate LLM miscalibration?

When their confidences were calibrated with traditional post-hoc techniques and sufficient data, all LLMs saw substantial reduction in Flex-ECE in all tasks. Histogram binning and isotonic regression were nearly equally effective in improving calibration, reducing the average Flex-ECE across models and datasets by 23.5 percentage points (for histogram binning) and 23.6 percentage points (for isotonic regression) when using 1000 calibration examples.

As shown in Table 4, using a larger number of calibration examples consistently led to larger improvements in calibration. Though using only 10 examples for calibration decreased the mean Flex-ECE for most models, we found the pattern of improvement to be erratic, with many models seeing an increase for several datasets. We found as few as 100 examples proved sufficient to provide a large and consistent reduction in Flex-ECE, with the average Flex-ECE falling by over 75% for all models other than Medicine-Llama3-8B and MedLLaMA-13B.

When a large enough number of calibration examples were used, the average Flex-ECE varied very little between tasks. For example, Flex-ECE ranged from 1.9% to 5.6% (averaged over models) for isotonic regression calibrated with a maximum of 500 examples, with QA standing out as being worse calibrated than most other tasks, having a Flex-ECE of 4.2% (though calibration for sentence similarity was technically

worse, BIOSSES has far few training examples than other datasets and so cannot be easily compared).

Discussion

Key takeaways

LLM calibration is very poor in BioNLP: The main message from the results is that the current state-of-the-art methods for confidence estimation in LLMs are not trustworthy enough for real-world BioNLP solutions. The best mean Flex-ECE (Medicine-Llama3-8B, 29.8%) is not a comfortable error margin; in practice, this means that LLM confidence in BioNLP is approximately 30% off the target at best. For comparison, the worst ECE scores in Guo et al.⁵ also for NLP tasks, albeit non-BioNLP, were approximately 8%, far below the best mean Flex-ECE we found in this BioNLP-specific study.

LLMs tend to be overconfident: The models exhibited general overconfidence (see summary in Table 2 and reliability diagrams in Supplementary File 3 for details) in their answers when questioned. This inability to exhibit uncertainty is a major blocker to the trustworthiness and therefore usability of these models, regardless of accuracy.

No “one size fits all” approach: We could not establish many reliable patterns between calibration and LLM families or confidence assignment strategies, which suggests that a tailored approach to assigning confidence scores to LLM responses is required in real-world, biomedical applications. The only clear pattern we observed is that Self-consistency was consistently more reliable than other strategies (lowest mean Flex-ECE in 11/13 tasks).

Post-hoc calibration helps: As a result, users are warned to place the necessary *guardrails* around LLMs when building LLM-backed, BioNLP solutions that rely on confidence estimation. Traditional post-hoc calibration techniques proved promising: despite being low-cost, they drastically reduced calibration errors, thereby increasing the degree of reliability of the confidence estimations we investigated in this study.

Known limitations

Verbal confidence elicitation: Because LLMs are highly susceptible to the wordings of prompts,^{31–33} it may be that the prompts to elicit verbal confidence estimations can be improved. Techniques such as *Chain-of-Thought*³⁴ and *p-tuning*³⁵ have proven useful to increase both LLM performance and reliability.

Table 4. Mean calibrated Flex-ECE scores by model, maximum number of examples, and calibration method.

# Examples	Calibration method	F-XXL	G-3	G-4	ML-13	ML-8	M-7	L-8	Y-34	Z-7
Uncalibrated		27.9	32.6	34.0	15.2	17.4	15.6	32.4	35.5	24.0
10	HB	21.2	20.3	17.1	21.6	29.0	0.4	25.6	28.5	9.7
	IR	17.0	15.6	15.7	10.2	15.1	0.4	15.0	20.0	7.4
50	HB	9.9	8.5	8.6	7.2	8.4	0.2	12.0	9.5	5.3
	IR	7.9	7.1	7.6	7.4	8.4	0.3	10.1	8.0	4.8
100	HB	6.2	4.9	6.4	6.9	6.0	0.2	7.6	7.2	2.8
	IR	5.5	4.7	6.3	6.1	6.4	0.2	7.0	7.0	3.1
500	HB	2.9	2.8	2.9	4.3	3.5	0.1	4.5	3.1	1.8
	IR	2.3	3.2	3.3	3.9	3.0	0.1	4.6	3.5	1.5
1000	HB	3.3	2.5	3.0	3.7	2.4	0.1	4.4	2.6	1.6
	IR	2.3	2.7	2.8	3.5	2.6	0.1	3.7	2.8	1.6

Confidences assignment strategy used is the average best strategy for each model in RQ1. Models are abbreviated as above. Calibration methods are abbreviated: HB = Histogram binning; IR = Isotonic regression. Bold values denote the best Flex-ECE score for each model and technique (IR and HB).

Verbal confidence parsing: Because LLMs can output any text, including unintelligible *gibberish*, verbal confidence estimation is often not feasible in an automated framework for confidence estimation, evaluation, and mitigation such as ours. Techniques such as *declarative prompting*^{36,37} that constraint LLM output may help avoid such scenarios.

Calibration is not utility: ECE (flex or not) can be a misleading quantification of reliability, because models can be very well-calibrated, yet have poor performance, ie, low utility. This is particularly true when confidence scores cluster around a certain range (ie, a *bucket*), a phenomenon dubbed *bucket collapse*.³⁸ We observed this mostly with the calibrated scores, ie, after-post-hoc calibration, when we ended up with most confidence scores *clumped* around the same low range of the LLM's performance, so that calibration could be improved. We urge users to couple (Flex-)ECE with the usual performance metrics (eg, F1 or accuracy) when judging the reliability of LLMs (and confidence estimation strategies).

Black-box confidence: All calibration and confidence estimation methods used in this study are designed to work with *black-box* access to an LLM, ie, users need not have access to the inner workings of LLMs to quantify and/or mitigate miscalibration. This was a deliberate decision in this study to allow any user to evaluate any LLM and/or apply this framework in their own work. Advanced users may wish to explore white-box confidence estimation techniques, such as *aggregating token probabilities*,³⁹ as they have shown some degree of success. Note that black-box use of closed-source LLMs such as GPT is the only way to use such LLMs by any user, advanced or not.

Conclusion

This study assesses the calibration of LLMs in BioNLP. The results corroborate previous studies that neural networks are poorly calibrated but highlights an even more alarming situation when networks are LLMs and the tasks are biomedically related: at best, LLMs are approximately 30% off the target in the scenarios we investigated.

In addition to the above findings, we also introduced Flex-ECE, a calibration metric that more realistically quantifies calibration of NLP models, and indeed any model that can be partially right or wrong. Flex-ECE is a key contribution of this study to the scientific community, as it opens path for future work to explore calibration methods that move away from binary correctness and towards a more continuous notion of correctness.

Lastly, we would like to reiterate that calibration is but a proxy to the trustworthiness of LLMs and indeed any neural network. Where possible, developers of LLM-backed applications should leverage as many metrics as possible to assess both the utility and trustworthiness of LLMs, especially in high stakes scenarios such as healthcare. Thus, the framework we here propose should *not* be viewed as the only approach to quantify the trustworthiness of LLMs in BioNLP, but one of the many useful approaches.

Author contributions

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Writing—review & editing), Matthew Garber (Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Writing—original draft, Writing—review & editing), James M Gwinnutt (Formal analysis, Validation, Writing—original draft, Writing—review & editing), Emaan Rashidi (Conceptualization, Writing—original draft, Writing—review & editing), Jwu-Hsuan Shantina Hwang (Software), William Gilmour (Conceptualization), Jay Nanavati (Conceptualization, Methodology, Writing—review & editing), Khaldoun Zine El Abidine (Writing—review & editing), and Christina DeFilippo Mack (Writing—original draft, Writing—review & editing)

Supplementary material

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Conflicts of interest

R.O. has stock in Arria NLG. J.N. has stock in Microsoft, AstraZeneca, Nvidia, Meta, and IQVIA. K.Z. has stock in IQVIA. C.M. has stock in AstraZeneca, Johnson & Johnson, MindMed, and IQVIA. R.O. was previously employed by Arria NLG. J.N. was previously employed by AstraZeneca.

Data availability

The data underlying this article are available in HuggingFace, at <https://huggingface.co/datasets>. The datasets were derived from sources in the public domain: BC2GM, BC5-chemical, BC5-disease, JNLPBA, and NCBI-disease can be found at <https://huggingface.co/datasets/bigbio/blurb>; EBM-PICO at https://huggingface.co/datasets/bigbio/ebm_pico; DDI at https://huggingface.co/datasets/bigbio/ddi_corpus; ChemProt at <https://huggingface.co/datasets/bigbio/chemprot>; GAD at <https://huggingface.co/datasets/bigbio/gad>; BIOSSES at <https://huggingface.co/datasets/bigbio/biosses>; HoC at https://huggingface.co/datasets/bigbio/hallmarks_of_cancer; BioASQ at <https://huggingface.co/datasets/Coldog2333/bioasq>; and PubMedQA at https://huggingface.co/datasets/bigbio/pubmed_qa.

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Research and Applications